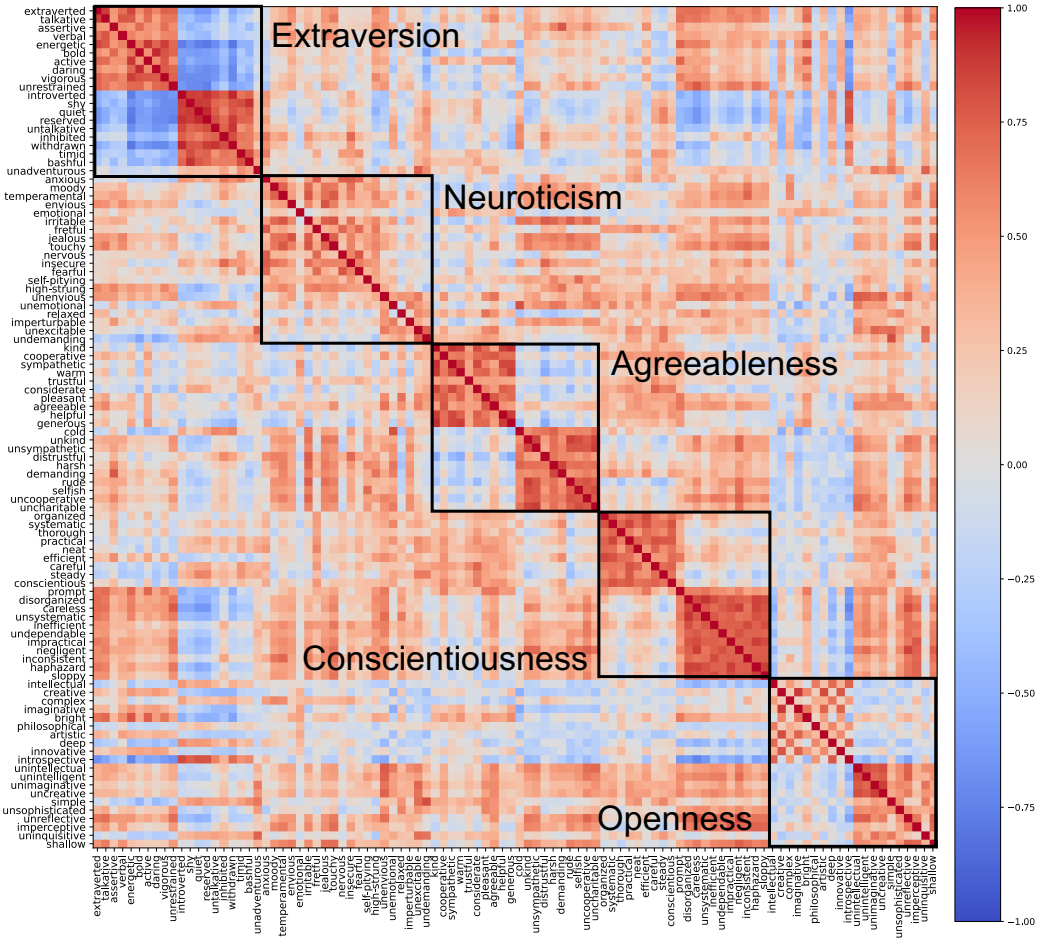

Figure 4: Visualization of correlation between trait adjectives. Datasets, models, and decoding parameters are identical to those of Figure 3. Five boxes with black edges indicate personality traits that adjectives belong to, drawn for visual aid. Trait adjectives that share Big Five trait show strong correlation, either positive or negative. We note that correlations between adjectives of different Big Five trait also show moderate level of correlation (e.g., 'introverted' and 'introspective'). This may imply that an adjective is related to several latent factors (instead of a single latent factdor) or that Big Five personality traits are not orthogonal.

Table 2: 100 unipolar trait descriptive adjectives grouped by Big Five traits and the pole. Pole refers to one end of a personality trait's continuum. Each of the five traits is considered a spectrum between two opposite extremes, or poles.

Trait	**Extraversion**		**Neuroticism**		**Agreeableness**	
Pole	(+)	(-)	(+)	(-)	(+)	(-)
Adjectives	extraverted	introverted	anxious	unenvious	kind	cold
	talkative	shy	moody	unemotional	cooperative	unkind
	assertive	quiet	temperamental	relaxed	sympathetic	unsympathetic
	verbal	reserved	envious	imperturbable	warm	distrustful
	energetic	untalkative	emotional	unexcitable	trustful	harsh
	bold	inhibited	irritable	undemanding	considerate	demanding
	active	withdrawn	fretful		pleasant	rude
	daring	timid	jealous		agreeable	selfish
	vigorous	bashful	touchy		helpful	uncooperative
	unrestrained	unadventurous	nervous		generous	uncharitable
			insecure			
			fearful			
			self-pitying			
			high-strung			

Trait	**Conscientiousness**		**Openness**	
Pole	(+)	(-)	(+)	(-)
Adjectives	organized	disorganized	intellectual	unintellectual
	systematic	careless	creative	unintelligent
	thorough	unsystematic	complex	unimaginative
	practical	inefficient	imaginative	uncreative
	neat	undependable	bright	simple
	efficient	impractical	philosophical	unsophisticated
	careful	negligent	artistic	unreflective
	steady	inconsistent	deep	imperceptive
	conscientious	haphazard	innovative	uninquisitive
	prompt	sloppy	introspective	shallow

A.3 Tokenization of Adjectives

As LLMs model next-token responses, a log-probability of an adjective is a sum of log-probabilities of tokens that consist the adjective. A single adjective is tokenized into multiple tokens. For example, an adjective 'sophisticated' is tokenized into four tokens with `Llama3` tokenizer: 's', 'oph', 'istic', and 'ated'. Therefore, to compute the log-probability for 'sophisticated' with the prompt (Figure 2), we compute four instances of log-probability: (1) log-probability of 's' given the prompt, (2) log-probability of 'oph' given the prompt appended with 's', (3) log-probability of 'istic' given the the prompt appended with 'soph', and (4) log-probability of 'ated' given the prompt appended with 'sophistic'. Four instances of log-probability are added to measure the log-probability for 'sophisticated'. Similar process is repeated for each trait adjectives.

A.4 Decoding Parameter: Temperature

Result in the main text is obtained with sampling temperature $T = 1.0$. We can compute the log-probability at different sampling temperatures with a single round of computation at $T = 1.0$, if we have access to log-probabilities for a complete set of vocabularies. This can be done with an efficient LLM serving system [20]. The process below describes the computation process at a token level.

Let N be a vocabulary size of a language model, W a list of conditioning tokens, and w_{target} a next token of which we want to compute log-probability. Our goal is to compute the log-probability $\text{LP}(w_{\text{target}}|W;T_{\text{t}})$ at a target temperature T_{t} given a full access to log-probabilities at an original temperature T_{o}, i.e. we know $\text{LP}(w_i|W;T_{\text{o}})$ for $i \in [1, N]$. We can use the following relation

$$\text{LP}(w_{\text{target}}|W;T_{\text{t}}) = \log\left(\frac{\exp\left(\frac{l_{\text{target}}}{T_{\text{t}}}\right)}{\sum_{i=1}^{N}\exp\left(\frac{l_i}{T_{\text{t}}}\right)}\right) = \log\left(\frac{1}{\sum_{i=1}^{N}\exp\left(\frac{l_i - l_{\text{target}}}{T_{\text{t}}}\right)}\right)$$

$$= \log\left(\frac{1}{\sum_{i=1}^{N}\exp\left(\frac{l_i - l_{\text{target}}}{T_{\text{o}}}\cdot\frac{T_{\text{o}}}{T_{\text{t}}}\right)}\right) = \log\left(\frac{1}{\sum_{i=1}^{N}\exp\left((\text{LP}(w_i|W;T_{\text{o}}) - \text{LP}(w_{\text{target}}|W;T_{\text{o}}))\cdot\frac{T_{\text{o}}}{T_{\text{t}}}\right)}\right)$$

where l_i represents a logit for the i-th token. Computation at an adjective level is straightforward by using token-level computations repeatedly for each token consisting an adjective (A.3).

B Dataset

Synthetic stories are adopted from the `PersonaLLM` dataset [16]. These stories were generated using `GPT-4-0613` by prompting with prescribed Big-Five personality traits (e.g., "You are a character who is extroverted, agreeable, conscientious, emotionally stable, and closed to experience.") along with an instruction "Please share a personal story in 800 words. Do not explicitly mention your personality traits in the story." The authors generated 10 stories for each of 32 combinations of binary Big-Five personality traits, and filtered out stories that explicitly contained trait-related lexicons. This process resulted in a total of 208 synthetic stories. We use these 208 stories as an input for log-probability measurement, and employ their binary Big-Five personality labels for the prediction accuracy measurement.